PRF-Reduced Sliding Spotlight SAR Imaging With Joint Sparse Representation Model

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***Abstract*— With the expansion of the surveillance area and resolution of spaceborne synthetic aperture radar (SAR) systems, the increasing amount of echo data requires further research on efficient imaging methods. The pulse repetition frequency (PRF) of traditional SAR needs to satisfy the Shannon–Nyquist sampling theory, while the PRF limits the system swath width, making it impossible to achieve wider illumination coverage. The reduction of the PRF can effectively increase the swath width, but it will cause severe azimuth ambiguity, decreasing the image quality. Several methods have been introduced for ambiguity suppression, but many of them become ineffective at lower PRFs. In this article, we propose a novel sparse imaging method for the spaceborne PRF-reduced sliding spotlight SAR. The proposed method considers the azimuth ambiguity term in the joint sparse imaging model, separately constrains the main imaging and azimuth ambiguity areas, and achieves azimuth ambiguity suppression by using compressive sensing (CS) technology. With its help, we can reduce the original PRF by up to half to double the swath width and enable high-precision sparse reconstruction of large-scale scenes. Compared with $L_{2,1}$-norm regularization-based algorithm, the proposed method shows the superior ambiguity suppression ability with less computational cost from PRF-reduced echo data. Experimental results on simulated raw data validate the proposed method.**

***Index Terms*— Azimuth ambiguity suppression, sliding spotlight, sparse SAR imaging, synthetic aperture radar (SAR).**

I. Introduction

SLIDING spotlight synthetic aperture radar (SAR) is a mode between stripmap and spotlight SAR, which has both the high-resolution spotlight and the wide swath of stripmap. The initial sliding spotlight SAR imaging method mainly focused on airborne sliding spotlight imaging, with low requirements on imaging methods concerning geometric relationship and resolution requirement [1], [2]. With the application of sliding spotlight mode in spaceborne SAR, high-precision imaging methods have emerged, which are mainly divided into subaperture and full-aperture-based methods. Since the designed pulse repetition frequency (PRF) of the sliding spotlight SAR system is typically lower than the total azimuth bandwidth, the collected echo data exhibits azimuth spectrum folding in the range–Doppler domain. The subaperture-based methods exploit aperture division to restore spectrum within the subaperture echo data [3], [4], whose representative algorithm is the baseband azimuth scaling (BAS) proposed by Prats et al. [4]. The steps of range processing of echo data in the BAS are based on the extended chirp scaling (ECS) [5], while subaperture partitioning is informed by the frequency scaling algorithm (FSA) suitable for spotlight SAR [6]. Similarly, many full-aperture imaging methods have been presented. Lanari et al. [7] proposed a full-aperture imaging method suitable for spotlight SAR. On this basis, Lanari et al. [1] introduced a novel imaging method, which removes the spectrum folding in the range–Doppler domain through the spectral analysis (SPECAN) in the first step. Sun et al. [8] and Zhou et al. [9] proposed a full-aperture imaging method to process the echoes of the sliding spotlight and terrain observation by progressive scans (TOPS) modes. Then, He et al. [10] considered the curvature of spaceborne SAR orbits and realized an ultrahigh resolution imaging of sliding spotlight. In addition, several methods have also been proposed to solve the squint [11], [12], [13] and multichannel [14], [15] sliding spotlight SAR imaging problems.

In typical cases, although the designed PRF of the system is lower than the total azimuth bandwidth, it will be higher than the azimuth beam bandwidth, thus enabling the above-mentioned imaging methods to be implemented. However, to further expand the illumination area, the PRF broke through the limitation of the Shannon–Nyquist sampling theory, which is referred to as reduced PRF. The imaging results by the aforementioned methods from the echo data with reduced PRF have noticeable azimuth ambiguity, which decreases the image quality. To solve this problem, Di Martino et al. [16] introduced a new method for mitigating azimuth ambiguity for stripmap SAR, which suppresses the ambiguities of antenna sidelobes by constructing two asymmetric filters. Chen et al. [17] proposed an ambiguity reconstruction and suppression framework based on the designed azimuth ambiguity factor. In the same year,

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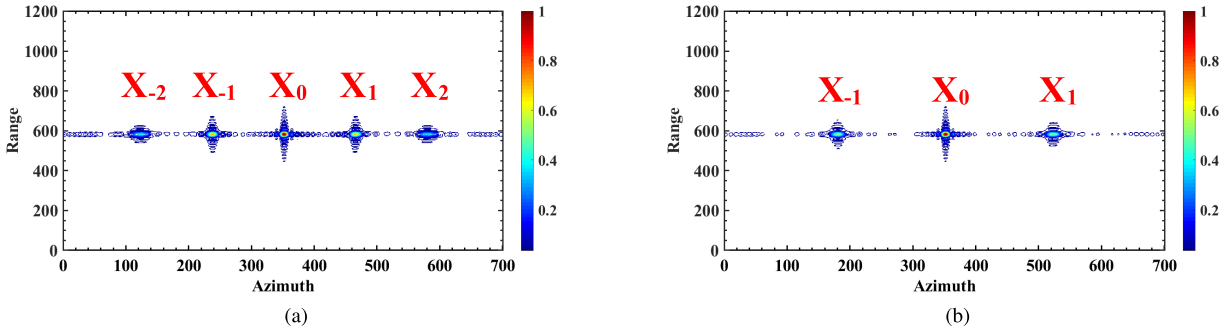


Fig. 1. Imaging results of a single-point target with different PRFs. (a) $k = 1/4$. (b) $k = 1/2$.

a new preprocessing method was demonstrated to suppress the residual ambiguity of multichannel SAR strong point-like targets [18]. Then, Long et al. [19] and [20], used filtering and azimuth ambiguity-to-signal ratio (AASR) estimation to decrease the effect of ambiguity on recovered images. Azimuth ambiguity can be suppressed by reversible signal processing [21] or by changing transmitted pulse sequence [22]. However, severe spectral aliasing renders filtering ineffective in restoring the targets while suppressing azimuth ambiguity with low PRFs. To address this issue and improve the imaging quality, compressive sensing (CS) technology has been introduced into SAR imaging by solving the $L_p(0 \leq p \leq 2)$-norm regularization problems, such as the least absolute shrinkage and selection operator (Lasso). The typical solution of the Lasso problem in sparse SAR imaging can suppress noise and sidelobes and hence improve the imaging quality. Xu et al. [23] achieved multichannel sliding spotlight SAR sparse imaging based on the Lasso model. However, it fails to achieve azimuth ambiguity suppression in the scene reconstruction. Yuan and Lin [24] and Yuan et al. [25] proposed the Group Lasso model, which grouped data based on their relationships and achieved sparse suppression at the group level. In 2011, on the basis of Lasso and Group Lasso models, Sprechmann et al. [26] proposed a Hierarchical Lasso (HiLasso) model and a Collaborative Hierarchical (C-HiLasso) model based on matrices, which constrains the data not only at the group level but also at the element level. The selection of regularization parameters in the proposed model can be based on [27] and [28]. The HiLasso model has been applied in various fields, for example, communication, hyperspectral analysis, and image classification [29], [30], [31], [32], [33], [34], [35], [36]. In SAR imaging, Qi et al. [37] discussed a range ambiguity suppression method combining matched filter and sparse reconstruction. Liu et al. [38] introduced a multichannel SAR imaging method for nonuniform sampling conditions inspired by $L_{2,1}$-norm regularization. Bi et al. [39] proposed an $L_{2,1}$-norm regularization-based sparse SAR imaging method to suppress the azimuth ambiguity caused by the periodic block sampling of spaceborne echo data. Then, Zhou et al. [40] utilized this method to achieve the suppression of low azimuth ambiguity for sliding spotlight SAR. However, when the azimuth ambiguity is severe with the further decrease of PRFs, the $L_{2,1}$-norm regularization-based sparse SAR imaging method is no longer effective in suppressing the ghosts caused by azimuth ambiguity.

Therefore, in this article, we propose a novel PRF-reduced sliding spotlight SAR sparse imaging method. In the proposed method, we first divide the image into the main and the ambiguity areas, then construct different matching operators of these areas to obtain a sparse representation model for unambiguous SAR imaging. During the algorithm iteration, different matching operators will focus on their corresponding regions. For example, for an ambiguous area, its matching operator will enhance the part that truly belongs to it. By comparing the changes in the images, the main imaging area can be better distinguished from the ambiguity ones, and the residual ambiguity energy of the main imaging area can be further weakened by subtracting the echoes of the ambiguity areas from the total echo data. Finally, the unambiguous sparse SAR image of the considered scene can be reconstructed by solving the mixed-norm optimization problem. Compared with matched filtering (MF)-based algorithms, the proposed method improves the signal-to-noise ratio (SNR) of the reconstructed image. While compared to the typical $L_{2,1}$-norm regularization-based method, it shows better ambiguity suppression ability with less computational complexity.

This article is organized as follows. Section II briefly introduces the HiLasso model-based sparse recovery algorithm. In Section III, the joint sparse representation model-based sparse SAR imaging method is introduced in detail from the imaging model, implementation algorithm, to computational complexity analysis. In Section IV, the experiments based on simulated raw data are presented to verify the proposed method. Finally, conclusions are drawn in Section V.

II. $L_{2,1}$-Norm Regularization-Based Sparse SAR Imaging

In this section, the $L_{2,1}$-norm regularization-based unambiguous sparse SAR imaging method is introduced, which is based on the HiLasso model [26]. In addition, the factors causing image quality degradation of the method with reduced PRF are analyzed. Normally, PRF satisfies the Shannon–Nyquist sampling theory. After further reducing PRF, the new PRF f_a' has the following relationship with the original one f_a, that is,

$$f_a' = k \cdot f_a, \quad 0 < k < 1 \tag{1}$$

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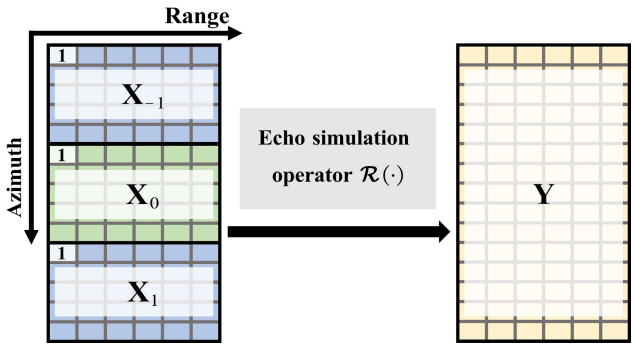


Fig. 2. Sparsity imaging model based on the echo simulation operator.

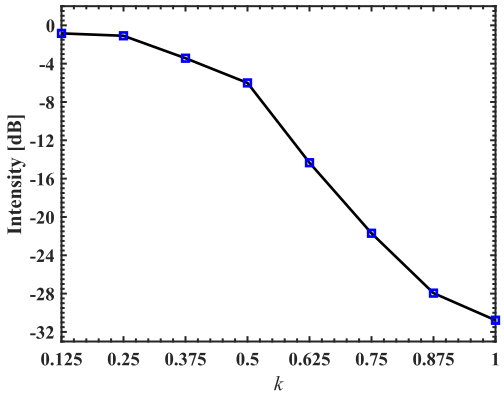


Fig. 3. Intensity change of the first ambiguity area of the point target with the reduction factor k.

where k is the reduction factor. In general, the oversampling rate of f_a is $1.1 \sim 1.2$. f_a', which does not satisfy the Shannon–Nyquist sampling theory, is named as reduced PRF.

A. Imaging Model

Let $\mathbf{Y}$ denote the 2-D echo data, and $\mathbf{X}$ denote the backscattering coefficient of the observed scene. Let $\mathbf{X}_i (i \in \mathbb{Z}$ and $i \neq 0)$ denote the azimuth ambiguity area in $\mathbf{X}$, and $\mathbf{X}_0$ denote the main imaging area which is the correct image in $\mathbf{X}$. Usually, azimuth ambiguity areas appear in pairs and are distributed on both sides of the imaging area. In this article, as shown in Fig. 1, it is named according to the distance between the azimuth ambiguity area and the main imaging area. Fig. 1 shows the imaging results of a single-point target from the data collected by different PRFs. From Fig. 1(a), it is seen that there is more than one set of azimuth ambiguity areas with different intensities. In general, the closer the ambiguity area is to the main imaging area, the greater its intensity. Thus, the image quality is mainly affected by the first azimuth ambiguity areas $\mathbf{X}_{-1}$ and $\mathbf{X}_1$ [see Fig. 1(b)]. Therefore, the discussed methods mainly focus on the suppression of $\mathbf{X}_{-1}$ and $\mathbf{X}_1$, although the methods can suppress all azimuth ambiguity areas theoretically. Since the azimuth ambiguity and the main imaging areas share the same support, we can group $\mathbf{X}$ and apply the sparse constraints to $\mathbf{X}$ by solving the HiLasso problem.

Considering the azimuth ambiguity terms, the sparse SAR imaging model can be rewritten as [38]

$$\mathbf{Y} = \mathcal{R}_0(\mathbf{X}_0) + \sum_i \mathcal{R}_i(\mathbf{X}_i) + \mathbf{N} \tag{2}$$

where $\mathcal{R}_0$ and $\mathcal{R}_i$ are the echo simulation operator of main imaging and the ith azimuth ambiguity areas, respectively, and $\mathbf{N}$ is the noise matrix. By analogy with the sparse pattern based on the observation matrix in [26], Fig. 2 shows the echo simulation operator-based sparse pattern. $\mathbf{X}$ is divided into $q (1 < q < Q)$ groups, with Q being the number of elements in $\mathbf{X}$. The element in $\mathbf{X}_0$ and the elements at the corresponding positions in $\mathbf{X}_1$ and $\mathbf{X}_{-1}$ form a group, that is, each group contains three elements. For example, in Fig. 2, the elements on the positions marked with 1 form a group. The approximated observation-based sparse SAR imaging model via HiLasso is depicted in (3), as shown at the bottom of the next page, where β_1 is the regularization parameter of $\mathbf{X}$ to represent the group sparsity of the whole scene and β_2 is the regularization parameter of $\mathbf{X}_0$ to represent the sparsity of main imaging area. HiLasso model further suppresses the residual ambiguity energy in the imaging area by constraining $\mathbf{X}$ at both the group level and the element level. The specific iterative procedure of model in (3) is detailed in [40].

B. Applicability

In spaceborne sliding spotlight SAR, the ability of the $L_{2,1}$-norm regularization-based method to suppress azimuth ambiguity deteriorates with the decrease of PRFs. There are two main reasons.

1) Low PRF: When the PRF satisfies the Shannon–Nyquist sampling theory, spectrum aliasing is not severe. The $L_{2,1}$-norm regularization-based method can weaken the azimuth ambiguity while keeping weak targets at low ghost intensity. However, when the echo spectrum is severely aliased due to low PRFs, this method will not be able to achieve effective ambiguity suppression. To further illustrate this point, single-point target imaging experiments are conducted with different k values. The amplitude of the point target is 1, which means that its normalized intensity is 0 dB. Fig. 3 shows the intensity change of the first ambiguity area of the point target with the reduction factor k. It is found that as the k value decreases, the ambiguity becomes more severe. Generally, the PRF reduction is not less than 75% of the original PRF and the intensity of the first ambiguity area is lower than the intensity of the first sidelobe of the target. However, when k drops to $1/2$, the intensity of the first ambiguity area increases to higher than -10 dB and the ghosts is unable to be eliminated by using the $L_{2,1}$-norm regularization-based method for ambiguity suppression.

2) Ambiguity Feature: Since the echo simulation operator is the inverse procedure of the MF-based algorithm [39], the selection of the used MF-based algorithm affects the performance of sparse imaging. In most sliding spotlight modes, to reduce computational complexity, the designed PRF is much lower than the total azimuth bandwidth and slightly higher than the azimuth beam bandwidth [23]. Unlike the stripmap mode, the MF-based imaging algorithms of sliding spotlight usually require first solving the azimuth spectral

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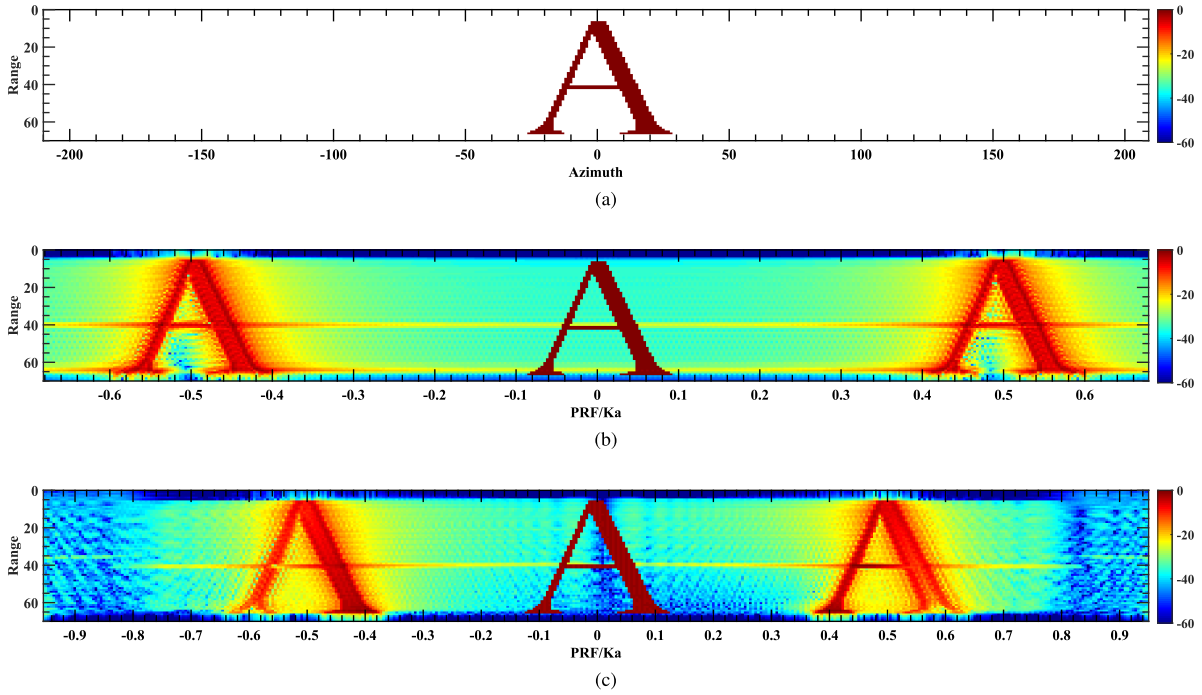


Fig. 4. Different features of the azimuth ambiguity between stripmap and sliding spotlight modes. (a) Simulated scene. (b) MF-recovered image of the stripmap mode with $k = 1/2$. (c) MF-recovered image of the sliding spotlight mode with $k = 1/2$.

folding, whether the algorithm is based on full-aperture processing [8] or subaperture division [4]. Due to the difference in the support, the inherent aliasing in the azimuth spectrum leads to the reduction of the effectiveness of the $L_{2,1}$-norm regularization-based method in ambiguity suppression. Fig. 4 shows the different features of the azimuth ambiguity between stripmap and sliding spotlight modes. It is seen that compared with the stripmap-based result, the imaging result of the azimuth ambiguity area in the sliding spotlight mode is more defocused, and the support is more different from the main imaging area, which makes the consistent support-based method less effective for azimuth ambiguity suppression.

Based on the above reasons, the $L_{2,1}$-norm regularization-based sparse SAR imaging method is not applicable for the azimuth ambiguity suppression in the sliding spotlight mode with reduced PRFs. In this article, inspired by the mixed-norm-based method, we divide the image into the main imaging area and the ambiguity areas and construct the joint sparse representation model to achieve azimuth ambiguity suppression with reduced PRFs. The $L_{2,1}$-norm-based method constrains $\mathbf{X}$ and $\mathbf{X}_0$. In comparison, the proposed method constrains $\mathbf{X}_0$ and $\mathbf{X}_i$, respectively, without group-level constraints.

III. Joint Sparse Representation Model-Based Unambiguous Sparse SAR Imaging

A. Model

In the data acquisition, the reduction of PRFs is equivalent to the uniform downsampling of the data collected by the original PRF satisfying the sampling theory. Therefore, in this article, we equate the data obtained with the reduced PRF to the uniform sampling of the data collected at twice the PRF value and write $\mathbf{Y}$ as

$$\mathbf{Y} = \mathbf{Y}_{\text{oral}} \circ \mathbf{H} \tag{4}$$

where $\mathbf{Y}_{\text{oral}}$ is the 2-D echo data collected by twice the reduced PRF, $\mathbf{H}$ is the sampling matrix with alternating 0–1 in the azimuth direction, and $\circ$ is the Hadamard product operator. Then, the range–Doppler spectrum of $\mathbf{Y}$ can be expressed as

$$S_a'(f_a) = \frac{1}{2}S_a(f_a) + \frac{1}{4}\left(S_a\left(f_a - \frac{\text{PRF}}{2}\right) + S_a\left(f_a + \frac{\text{PRF}}{2}\right)\right) \tag{5}$$

where $S_a'(f_a)$ is the range–Doppler spectrum of $\mathbf{Y}$, $S_a(f_a)$ is the range–Doppler spectrum of $\mathbf{Y}_{\text{oral}}$, and f_a is the azimuth frequency. From (5), it is seen that the azimuth ambiguity is mainly decided by the last term on the right side. The azimuth frequency of $\mathbf{X}_1$ and $\mathbf{X}_{-1}$ are $f_a + \text{PRF}/2$ and $f_a - \text{PRF}/2$, respectively. Because $S_a(f_a - \text{PRF}/2) = S_a(f_a + \text{PRF}/2)$, $\mathbf{X}_1$ and $\mathbf{X}_{-1}$ are considered as a fused $\mathbf{X}_a$. Thus, we can rewrite the imaging model in (2) as

$$\mathbf{Y} = \mathcal{R}_0(\mathbf{X}_0) \circ \mathbf{H} + \mathcal{R}_a(\mathbf{X}_a) \circ \mathbf{H} + \mathbf{N} \tag{6}$$

where $\mathcal{R}_a(\cdot)$ is the echo simulation operator of $\mathbf{X}_a$. The sparse estimation of $\mathbf{X}_0$ can be recovered by solving the optimization

$$\hat{\mathbf{X}} = \underset{\mathbf{X}}{\text{argmin}}\left\{\left\|\mathbf{Y} - \mathcal{R}_0(\mathbf{X}_0) - \sum_i \mathcal{R}_i(\mathbf{X}_i)\right\|_{\text{F}}^2 + \beta_1\|\mathbf{X}\|_{2,1} + \beta_2\|\mathbf{X}_0\|_1\right\} \tag{3}$$

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Algorithm 1 Iterative Recovery for the Model in (23)

Input:

Echo data $\mathbf{Y}$; Operators $\mathcal{R}_0$, $\mathcal{G}_0$

Initial

$\mathbf{X}_0^{(0)} = \mathbf{0}$; $\mathbf{X}_a^{(0)} = \mathbf{0}$; $\mathbf{U}_0^{(0)} = \mathbf{0}$; $\mathbf{U}_a^{(0)} = \mathbf{0}$
Iterative parameter μ; Error parameter ε
Maximum iterative step $L_{\max}$

while $0 < l \leqslant L_{\max}$ and Resi $> \varepsilon$ **do**

Estimate residual

$$\Delta^{(l-1)} = \mathbf{Y} - \left(\mathcal{R}_0\left(\mathbf{X}_0^{(l-1)}\right) + \mathcal{R}_a\left(\mathbf{X}_a^{(l-1)}\right)\right) \circ \mathbf{H}$$

Update gradient

$$\mathbf{U}_0^{(l-1)} = \mathbf{X}_0^{(l-1)} + \mu \mathcal{G}_0\left(\Delta^{(l-1)}\right)$$

$$\mathbf{U}_a^{(l-1)} = \mathbf{X}_a^{(l-1)} + \mu \mathcal{G}_a\left(\Delta^{(l-1)}\right)$$

Update β_1 and β_2

$$\beta_1^{(l-1)} = \left|\mathbf{U}_0^{(l)}\right|_{K+1} / \mu$$

$$\beta_2^{(l-1)} = \left|\mathbf{U}_a^{(l)}\right|_{K+1} / \mu$$

Thresholding

$$\mathbf{X}_0^{(l)} = G_{1,\mu\beta_1}\left(\mathbf{U}_0^{(l)}\right)$$

$$\mathbf{X}_a^{(l)} = G_{1,\mu\beta_2}\left(\mathbf{U}_a^{(l)}\right)$$

$$\mathbf{X}_0^{(l)} = \mathbf{X}_0^{(l)} \cdot \operatorname{sign}\left(\frac{\left|\mathbf{X}_0^{(l)}\right|}{\max\left(\left|\mathbf{X}_0^{(l)}\right|\right)} - \frac{\left|\mathbf{X}_a^{(l)}\right|}{\max\left(\left|\mathbf{X}_a^{(l)}\right|\right)} - \delta\right)$$

Calculate the residual of recovered image

$$\text{Resi} = \left\|\mathbf{X}_0^{(l)} - \mathbf{X}_0^{(l-1)}\right\|_{\mathrm{F}}$$

$$l = l + 1$$

end while

Output:

Recovered sparse image $\hat{X} = \mathbf{X}_0^{(l)}$

problem in (7), as shown at the bottom of the page [31], where β_1 is the regularization parameter of $\mathbf{X}_0$, and β_2 is the regularization parameter of $\mathbf{X}_a$.

B. Echo Simulation Operator Construction

The MF-based imaging algorithm is mainly divided into three steps: azimuth preprocessing, range compression and range cell migration correction (RCMC), and azimuth compression. In the azimuth preprocessing, spectral analysis (SPECAN) is used to address aliasing in the range–Doppler domain. Different from [8], the hyperbolic model of the distance equation is retained. Therefore, in sliding spotlight SAR, the echo of the target at (t_0, r_0) can be expressed as

$$\begin{aligned} s(t_a, t_r; t_0, r_0) = {} & \operatorname{rect}\left(\frac{t_r - 2R(t_a; t_0, r_0)/c}{T_r}\right) \\ & \cdot \operatorname{rect}\left(\frac{t_a - t_0}{T_{\text{slide}}}\right) \\ & \cdot \exp\left\{-j\frac{4\pi R(t_a; t_0, r_0)}{\lambda}\right\} \\ & \cdot \exp\left\{-j\pi K_r\left(t_r - \frac{2R(t_a; t_0, r_0)}{c}\right)^2\right\} \end{aligned} \tag{8}$$

where t_a and t_r are the azimuth and range times, r_0 is the shortest slant range from the target to the platform, t_0 is the beam center crossing time, c is the speed of light, λ is the carrier wavelength, T_r is the range pulse duration, K_r is the range frequency modulation rate, T_{slide} is the target illumination time, and $R(t_a; t_0, r_0)$ is the instantaneous slant range with

$$R(t_a; t_0, r_0) = \sqrt{{r_0}^2 + (t_a - t_0)^2 {V_r}^2} \tag{9}$$

where V_r is the effective velocity. The azimuth preprocessing can be simply written as

$$s_1\left(t_a', t_r\right) = \mathscr{F}_a[s(t_a, t_r; t_0, r_0) \circ \mathbf{H}_{\text{ref}}] \circ \mathbf{H}'_{\text{ref}} \tag{10}$$

where $\mathscr{F}_a[\cdot]$ is the azimuth Fourier transform operator, and t_a' is the new azimuth time after the azimuth preprocessing. The detailed expressions of $\mathbf{H}_{\text{ref}}$ and $\mathbf{H}'_{\text{ref}}$ can be found in [1]. Since the hyperbolic model of the distance equation is retained, the range compression, RCMC, and secondary range compression (SRC) operations ($\mathbf{H}_1$, $\mathbf{H}_2$, and $\mathbf{H}_3$) are performed by using ECS [5], that is,

$$\begin{aligned} & S_{a2}\left(f_a', t_r\right) \\ & \quad = \mathscr{F}_r^{-1}\left[\mathscr{F}_r\left[\mathscr{F}_a\left[s_1\left(t_a', t_r\right)\right] \circ \mathbf{H}_0 \circ \mathbf{H}_1\right] \circ \mathbf{H}_2\right] \circ \mathbf{H}_3 \end{aligned} \tag{11}$$

where $\mathscr{F}_r[\cdot]$ and $\mathscr{F}_r^{-1}[\cdot]$ are the Fourier transform and inverse Fourier transform operators in the range direction, respectively, f_a' is the azimuth frequency corresponding to t_a', and $\mathbf{H}_0$ is the phase compensation operation, which can be expressed as

$$\mathbf{H}_0\left(f_a', t_r\right) = \exp\left\{j2\pi \frac{\lambda R_{\text{rot}}}{4V_r^2} {f_a'}^2\right\} \tag{12}$$

where R_{rot} is the slant range of rotation point. Meanwhile, the azimuth compression also needs to be modified. SPECAN is

$$\hat{\mathbf{X}} = \underset{\mathbf{X}}{\operatorname{argmin}}\left\{\|\mathbf{Y} - \mathcal{R}_0(\mathbf{X}_0) \circ \mathbf{H} - \mathcal{R}_a(\mathbf{X}_a) \circ \mathbf{H}\|_{\mathrm{F}}^2 + \beta_1 \|\mathbf{X}_0\|_1 + \beta_2 \|\mathbf{X}_a\|_1\right\} \tag{7}$$

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TABLE I
SIMULATED PARAMETERS OF THE SLIDING SPOTLIGHT MODE

Parameter	Value
Orbit altitude [km]	600
Platform velocity [m/s]	7600
Middle incident angle [deg]	38.41
Wavelength [m]	0.031
Pulse duration [μs]	20
Signal bandwidth [MHz]	150
Reduced PRF [Hz]	1650
Antenna [m]	5.40
Azimuth resolution [m]	1.0
Maximum steering angle [deg]	± 3.5

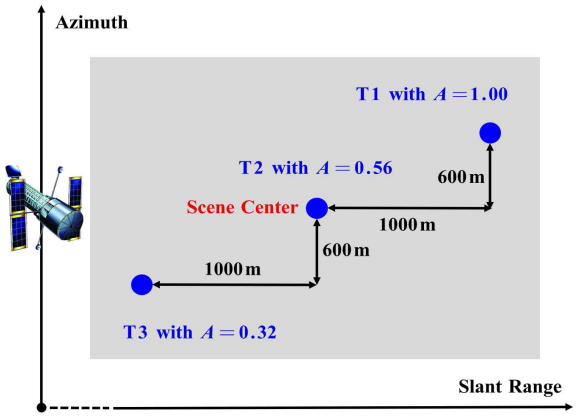


Fig. 5. Target location in the experiments based on simulated data.

used to realize azimuth compression, that is,

$$s_3(t_a', t_r) = \mathscr{F}_a\left[\mathscr{F}_a^{-1}\left[S_{a2}(f_a', t_r) \circ \mathbf{H}_4\right] \circ \mathbf{H}_5\right] \circ \mathbf{H}_6. \quad (13)$$

The first step of SPECAN is the hyperbolic azimuth phase removal and replace it with a purely quadratic phase shape by

$$\mathbf{H}_4(f_a', t_r) = \exp\left\{-j4\pi \frac{1-\sqrt{1-\frac{\lambda^2 f_a'^2}{4V_r^2}}}{\lambda} r\right\} \cdot \exp\left\{j\pi \frac{\lambda(R_{\text{rot}} - R_c)}{2V_r^2} f_a'^2\right\} \quad (14)$$

where R_c is the slant range of scene center and $r = (c \cdot t_r)/2$. $\mathbf{H}_5$ can be expressed as

$$\mathbf{H}_5(t_a', t_r) = \exp\left\{j\pi \frac{2V_r^2}{\lambda(R_{\text{rot}} - R_c)} t_a'^2\right\}. \quad (15)$$

For phase preservation, we need to multiply

$$\mathbf{H}_6(t_a'', t_r) = \exp\left\{j\pi \frac{\lambda(R_{\text{rot}} - r)}{2V_r^2} t_a''^2\right\} \quad (16)$$

where t_a'' is the azimuth time after SPECAN. In summary, the MF imaging operator $\mathcal{G}(\cdot)$ of sliding spotlight SAR is depicted in (17), as shown at the bottom of the next page, whose echo simulation operator $\mathcal{R}(\cdot)$ is shown in (18), as shown at the bottom of the next page, with $(\cdot)^*$ being the conjugate operator. The construction of $\mathcal{R}_0(\cdot)$ and $\mathcal{R}_a(\cdot)$ are identical to $\mathcal{R}(\cdot)$. The difference is the azimuth frequency values. According to (5), the relationship of $\mathcal{R}_0$ and $\mathcal{R}_a$ can be further expressed as

$$\mathcal{R}_a(\mathbf{X}_a) = \mathscr{F}_a^{-1}\left[\text{circshift}\left[\mathscr{F}_a[\mathcal{R}_0(\mathbf{X}_0)], N_a/2\right]\right] \quad (19)$$

where N_a is the number of elements of $\mathbf{X}_0$ in the azimuth direction. $A = \text{circshift}[A, M]$ denotes the circularly shifts the elements in A by M positions along the azimuth direction.

C. Iterative Recovery

For the model in (7), the threshold iterative solution sequence is shown in (20), as shown at the bottom of the next page, where μ is the parameter that controls the algorithm convergence speed and satisfies the relationship

$$0 < \mu < \min\left\{\|\mathcal{R}_0(\mathbf{X}_0)\|_{\text{F}}^2, \|\mathcal{R}_a(\mathbf{X}_a)\|_{\text{F}}^2\right\}. \quad (21)$$

$G_{1,\mu,\beta_2}(\mathbf{U}_a^{(l)})$ and $G_{1,\mu,\beta_2}(\mathbf{U}_a^{(l)})$ are the traditional iterative soft thresholding functions [41]. In the iteration, since $\mathcal{R}_0(\cdot)$ and $\mathcal{G}_0(\cdot)$ are the matching operators of $\mathbf{X}_0$, the part that truly belongs to $\mathbf{X}_0$ is focused. The true parts of ambiguity areas are focused similarly to $\mathbf{X}_0$. However, $\mathbf{X}_0$ and $\mathbf{X}_a$ have inconsistent intensity ranges, so we first normalize $\mathbf{X}_0$ and $\mathbf{X}_a$. Then, by comparing the normalized amplitude of $\mathbf{X}_0$ and $\mathbf{X}_a$, we can distinguish between the main and the ambiguity areas and further suppress the residual azimuth ambiguity energy in the main imaging area. This processing can be written as

$$\mathbf{X}_0 = \mathbf{X}_0 \cdot \text{sign}\left(\frac{|\mathbf{X}_0|}{\max(|\mathbf{X}_0|)} - \frac{|\mathbf{X}_a|}{\max(|\mathbf{X}_a|)} - \delta\right) \quad (22)$$

where δ is the discrimination threshold. The detailed iterative procedure to solve the optimization problem in (20) is listed in Algorithm 1.

D. Computational Complexity Analysis

For clarity, the $L_{2,1}$-norm regularization-based sparse imaging method discussed in Section II is named $L_{2,1}$-Sp. Let I_1 and I_2 denote the number of required iterative steps of accurate recovery for $L_{2,1}$-Sp and the proposed method. Let P and $Q = N_a \times N_r$ represent the number of elements in $\mathbf{Y}$ and $\mathbf{X}$, respectively. For each iteration of sparse imaging methods, their computation includes two main parts. The first one is the calculation of the echo simulation operator and the MF operator. The complexity of $L_{2,1}$-Sp is in the order of $\mathcal{O}(6P \log P)$ when we only consider the first azimuth ambiguity area in the scene reconstruction. In comparison, since $\mathcal{R}_a(\mathbf{X}_a)$ can obtain from $\mathcal{R}_0(\mathbf{X}_0)$, the computational cost of the proposed method is about $\mathcal{O}(2P \log P)$. After considering the complexity of the thresholding operation, the total computational complexity of $L_{2,1}$-Sp and the proposed method are $\mathcal{O}(6I_1 P \log P + 4Q)$ [39] and $\mathcal{O}(2I_2 P \log P + 3Q)$, respectively. Due to the reduction of the number of operators in each iteration, the computational cost of the proposed method is less than that $L_{2,1}$-Sp, but more than that of the traditional L_1-norm regularization-based method.

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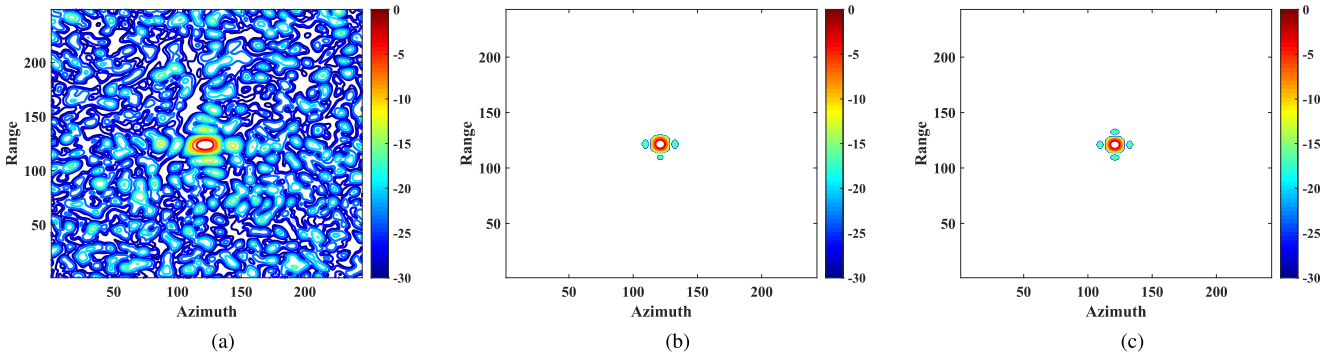


Fig. 6. Reconstructed images of T2 with reduced PRF by different methods. (a) MF. (b) $L_{2,1}$-Sp. (c) Proposed method.

IV. Experiments and Discussions

In this section, several experiments based on simulated data are used to verify the proposed method. The experimental parameters are listed in Table I.

A. Experiments Based on Simulated Data

Set three point targets T1, T2, and T3 with amplitudes of 1, 0.56, and 0.32, respectively, which means T1, T2, and T3 intensities are approximately 0, −5, and −10 dB, respectively. Their relative positions are shown in Fig. 5, where T2 is located at the center of the scene. The simulated echo data is collected by the reduced PRF ($k = 1/2$), with the SNR being −33 dB. Taking T2 as an example, the imaging results of different methods are shown in Fig. 6. It is found that the proposed method has a similar noise and sidelobe suppression ability as $L_{2,1}$-Sp, which is a common advantage of sparse imaging methods. Furthermore, target-to-background ratio (TBR) is used to quantitatively evaluate the noise and clutter suppression ability of different methods [39], that is,

$$\mathrm{TBR} \triangleq 20\log_{10}\left(\frac{\max\limits_{(p,q)\in\mathcal{I}}\left|\mathbf{X}_{(p,q)}\right|}{\frac{1}{Q_{\mathcal{B}}}\sum_{(p,q)\in\mathcal{B}}\left|\mathbf{X}_{(p,q)}\right|}\right) \tag{23}$$

where $\mathcal{I}$ represents the target area, $\mathcal{B}$ represents the background region around $\mathcal{I}$, and $Q_{\mathcal{B}}$ is the number of elements in $\mathcal{B}$. Table II shows the TBR values of the images recovered by different methods. Similar to Fig. 6, the proposed method shows excellent performance in improving the image quality.

Fig. 7 shows the reconstructed images by MF, $L_{2,1}$-Sp, and the proposed method imaging from the data collected by the reduced PRF ($k = 1/2$). In Fig. 7(a), the ambiguity intensity of T1, T2, and T3 are −6.01, −11.17, and −16.54 dB, respectively. From Fig. 7(b), it is seen that due to the high intensity of the azimuth ambiguity corresponding to T1, even if T3 is lost in the recovery, these ambiguities still cannot be suppressed. In comparison, the proposed method suppresses all ambiguous parts of T1 while retaining T3. Furthermore, AASR is used to quantitatively evaluate the ambiguity suppression capability of different methods, which is defined as

$$\mathrm{AASR} \triangleq 20\log_{10}\left(\frac{\frac{1}{Q_a}\sum_{(p,q)}\left|\mathbf{X}_{a(p,q)}\right|}{\frac{1}{Q_0}\sum_{(p,q)}\left|\mathbf{X}_{0(p,q)}\right|}\right) \tag{24}$$

where Q_0 and Q_a are the number of elements in $\mathbf{X}_0$ and $\mathbf{X}_a$, respectively. Table III shows the AASR values of the images recovered by different methods. Due to $L_{2,1}$-Sp loses T3 in the imaging, resulting in the failed reconstruction, its AASR value is not given for comparison. From Table III, it is seen that compared with MF and $L_{2,1}$-Sp, the proposed method can effectively suppress azimuth ambiguity, whose AASR value is increased by at least 14 dB.

B. Experiments Based on Real Scene Simulated Data

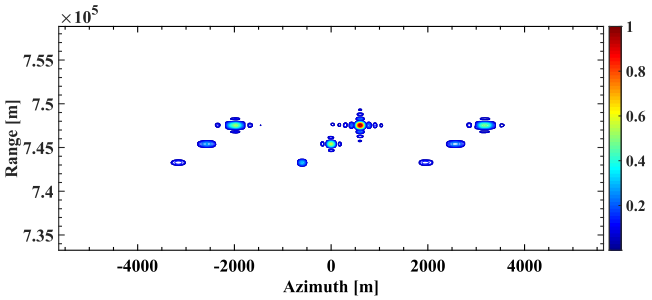
Since there is no publicly available sliding spotlight SAR echo data, in the experiments of this article, we simulated the echo data of the real-scene images in sliding spotlight mode using the parameters listed in Table I. Figs. 8 and 9 show the reconstructed images of the sea surface and island scenes by different methods from the simulated echo data of sliding spotlight SAR data with $k = 1/2$. For comparison, the

$$\mathcal{G}(\cdot) = \mathscr{F}_a\left[\mathscr{F}_a^{-1}\left[\mathscr{F}_r^{-1}\left[\mathscr{F}_r\left[\mathscr{F}_a\left[\mathscr{F}_a[\cdot\circ\mathbf{H}_{\mathrm{ref}}]\circ\mathbf{H}'_{\mathrm{ref}}\right]\circ\mathbf{H}_0\circ\mathbf{H}_1\right]\circ\mathbf{H}_2\right]\circ\mathbf{H}_3\circ\mathbf{H}_4\right]\circ\mathbf{H}_5\right]\circ\mathbf{H}_6 \tag{17}$$

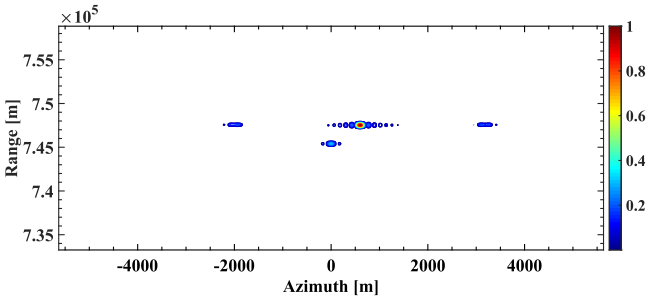
$$\mathcal{R}(\cdot) = \mathscr{F}_a^{-1}\left[\mathscr{F}_a^{-1}\left[\mathscr{F}_r^{-1}\left[\mathscr{F}_r\left[\mathscr{F}_a\left[\mathscr{F}_a^{-1}\left[\cdot\circ\mathbf{H}_6^*\right]\circ\mathbf{H}_5^*\right]\circ\mathbf{H}_4^*\circ\mathbf{H}_3^*\right]\circ\mathbf{H}_2^*\right]\circ\mathbf{H}_1^*\circ\mathbf{H}_0^*\right]\circ\left(\mathbf{H}'_{\mathrm{ref}}\right)^*\right]\mathbf{H}_{\mathrm{ref}}^* \tag{18}$$

$$\begin{aligned}\mathbf{X}^{(l)} &= \underset{\mathbf{X}}{\mathrm{argmin}}\left\{\left(\|\mathbf{X}_0-\mathbf{U}_0\|_{\mathrm{F}}^2+\|\mathbf{X}_a-\mathbf{U}_a\|_{\mathrm{F}}^2\right)+\beta_1\|\mathbf{X}_0\|_1+\beta_2\|\mathbf{X}_a\|_1\right\}\\ &= G_{1,\mu,\beta_1}\left(\mathbf{U}_0^{(l-1)}\right)+G_{1,\mu,\beta_2}\left(\mathbf{U}_a^{(l-1)}\right)\end{aligned} \tag{20}$$

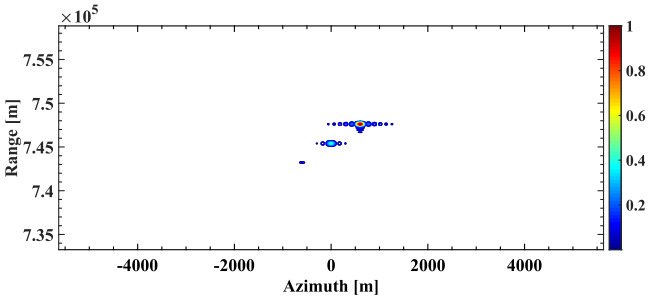
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(a)



(b)



(c)

Fig. 7. Reconstructed images of three simulated point targets with reduced PRF by different methods. (a) MF. (b) $L_{2,1}$-Sp. (c) Proposed method.

TABLE II
TBR VALUES OF POINT TARGETS IN THE RECOVERED IMAGES OF DIFFERENT METHODS [dB]

	T1	T2	T3
MF	15.83	15.43	15.98
$L_{2,1}$**-Sp**	34.20	40.18	37.85
The proposed method	37.35	41.33	38.20

MF-based images of surveillance regions from the data collected by the original PRF are depicted in Figs. 8(a) and 9(a). Similarly, the TBR values of the selected regions (marked by the yellow box) in Figs. 8 and 9 are also calculated for verification. Since the target of interest is no longer a point target, we have modified the definition of TBR in (32), that is,

$$\mathrm{TBR} \triangleq 20\log_{10}\left(\frac{\frac{1}{Q_m}\sum_{(p,q)\in\mathcal{I}}\left|\mathbf{X}_{(p,q)}\right|}{\frac{1}{Q_{\mathcal{B}}}\sum_{(p,q)\in\mathcal{B}}\left|\mathbf{X}_{(p,q)}\right|}\right) \tag{25}$$

TABLE III
AASR VALUES OF POINT TARGETS IN THE RECOVERED IMAGES OF DIFFERENT METHODS [dB]

	T1	T2	T3
MF	-6.60	-5.95	-6.56
$L_{2,1}$**-Sp**	-14.29	-29.35	/
The proposed method	-43.83	-44.32	-46.20

TABLE IV
TBR VALUES OF SELECTED AREAS IN THE RECOVERED IMAGES OF DIFFERENT METHODS [dB]

	T4	T7
MF method	21.99	23.32
$L_{2,1}$**-Sp**	30.24	33.03
The proposed method	38.58	36.60

TABLE V
AASR VALUES OF SELECTED AREAS IN THE RECOVERED IMAGES OF DIFFERENT METHODS [dB]

	T5	T6	T7
MF method	-0.25	-0.23	-0.41
$L_{2,1}$**-Sp**	-1.02	-1.31	-0.65
The proposed method	-12.63	-19.71	-28.60

where Q_m and is the number of elements in $\mathcal{I}$. Table IV shows the TBR values of the images recovered by different methods. From Figs. 8 and 9, and Table IV, it is seen that the proposed method has the similar noise and clutter suppression ability as $L_{2,1}$-Sp. Both of them can improve the SAR imaging performance compared to the MF-based method. However, the performance of the proposed method is better, and by observing T4 and T7, it can be found that the proposed method retains more target information than $L_{2,1}$-Sp while suppressing clutter. Table V shows the AASR values of T5, T6, and T7 regions in the recovered images of different methods. Similar to the point target simulations, it is seen that the proposed method shows excellent azimuth ambiguity suppression ability, which significantly improves the quality of sparse SAR images with reduced PRF. It is worth mentioning that the errors caused by nonideal $\mathcal{R}_0(\cdot)$ and $\mathcal{G}_0(\cdot)$ can affect the image quality since $\mathcal{R}_a(\cdot)$ and $\mathcal{G}_a(\cdot)$ are related to $\mathcal{R}_0(\cdot)$ and $\mathcal{G}_0(\cdot)$ in the constructed model of the proposed method. The degradation in image quality due to the errors is not considered in the experiments. The errors can be compensated for using other error estimation methods before employing the proposed method for ambiguity suppression.

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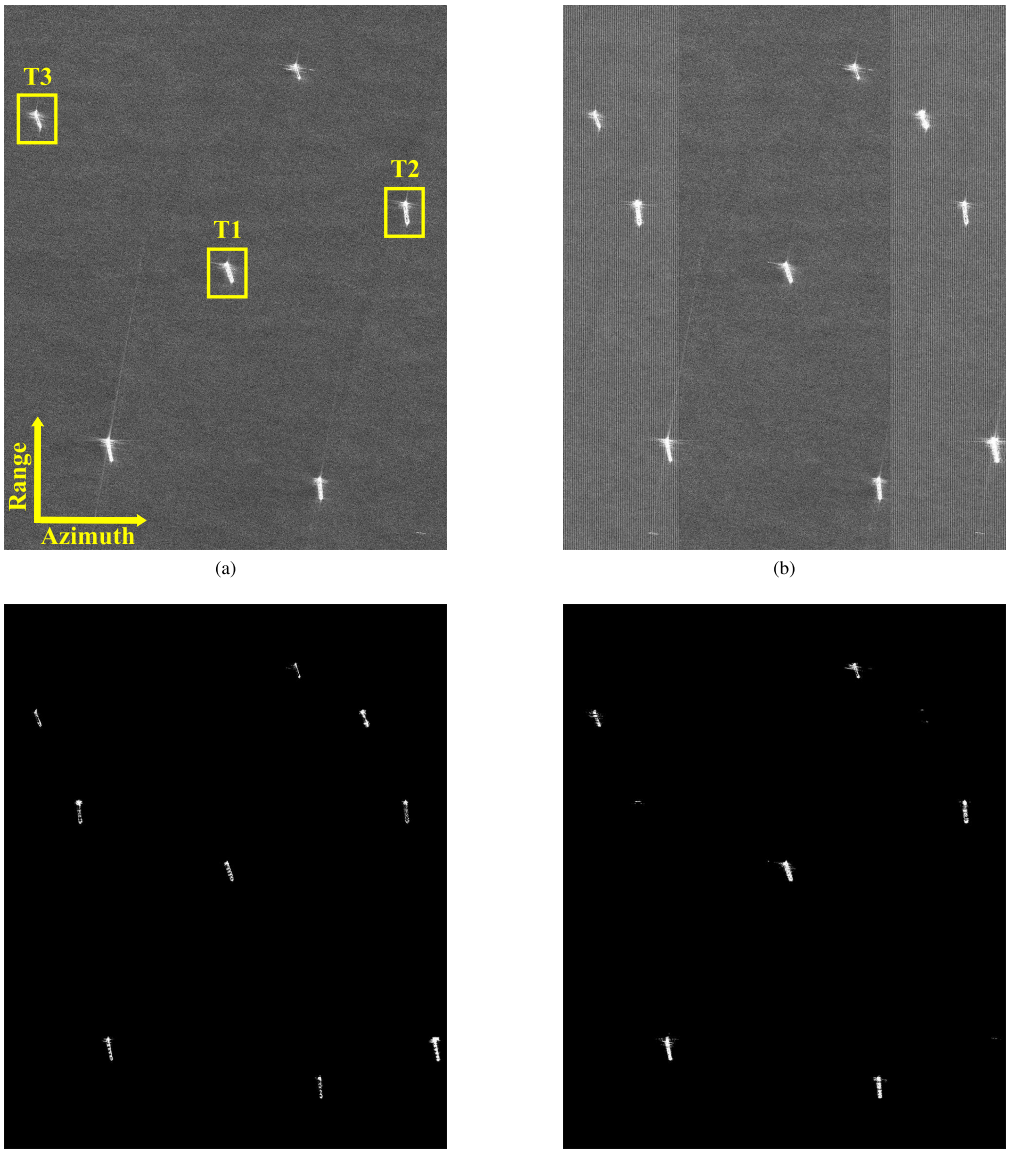


Fig. 8. Reconstructed images of sea surface area with reduced PRFs by different methods. (a) MF with original PRFs. (b) MF with reduced PRFs ($k = 1/2$). (c) $L_{2,1}$-Sp with reduced PRFs ($k = 1/2$). (d) Proposed method with reduced PRFs ($k = 1/2$).

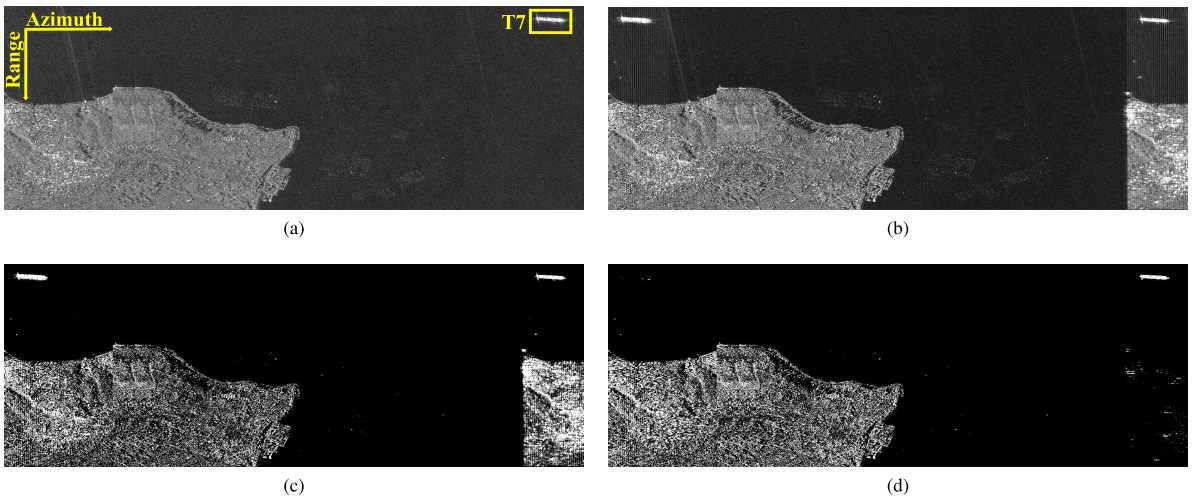


Fig. 9. Reconstructed images of island area with reduced PRF by different methods. (a) MF with original PRFs. (b) MF with reduced PRFs ($k = 1/2$). (c) $L_{2,1}$-Sp with reduced PRFs ($k = 1/2$). (d) Proposed method with reduced PRFs ($k = 1/2$).

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V. Conclusion

In this article, we propose a novel joint sparse representation model-based sparse imaging method for PRF-reduced sliding spotlight SAR data processing. In the proposed method, a joint sparse representation model is first constructed by considering the azimuth ambiguity terms caused by the PRF reduction. Then, by sparsely constraining the imaging and the azimuth ambiguity areas in the model, the residual azimuth ambiguity energy in the main imaging area and the ghosts caused by azimuth ambiguity are suppressed by solving a joint optimization problem. Compared to MF-based algorithms, the proposed method has the noise and clutter suppression ability as traditional sparse imaging methods. It can perform high-quality imaging of sparse scenes from the data collected by the reduced PRF, further improving the imaging performance of large-scale sparse scenes. Compared with the $L_{2,1}$-Sp algorithm, it shows superior ambiguity suppression ability and reduces the computational complexity dramatically. It is noted that similar to other sparse imaging methods, due to the limited number of samples used, the proposed method is more applicable for sparse scene recovery, such as a fleet of ships on the vast sea, islets, and so on.

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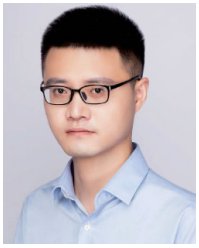
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